



SILVER OAK UNIVERSITY
Silver Oak College of Computer Application
Master of Science Data Science
Course Name: Information and Data Security
Course Code: DSM5061C
Semester: 1st

Prerequisite:

1. Basic Computer Fundamentals
2. Fundamentals of Information Technology
3. Basic Knowledge of Computer Networks and Internet Technologies
4. Familiarity with Data Handling and Database Concepts

Course Objectives:

1. To understand the fundamental concepts, principles, and challenges associated with information and data security in modern digital environments.
2. To develop an understanding of data privacy principles, data protection techniques, and secure data handling practices.
3. To explore security technologies and mechanisms used to protect data-driven systems, cloud environments, and digital infrastructure.
4. To understand emerging security challenges in data science, including AI-enabled security risks and privacy concerns.
5. To analyze governance, compliance, legal, ethical, and regulatory requirements related to information security and data protection.

Teaching Scheme:

Teaching & Learning Scheme (In Hrs Per Semester)									Total Credits
Classroom Instruction						Extended Learning		Total Hours	
L		T		P					
Cr	Hr	Cr	Hr	Cr	Hr	TW	SL	Hr	C
4	60	0	0	0	0	40	20	120	4

Contents:

Unit	Topics	Teaching Hours	% Weightage
1	Foundations of Information and Data Security Information Security Fundamentals: Introduction to Information and Data Security, Information Assets, Security Challenges in Data-Driven Organizations, Importance of Information Security. Core Security Principles: Confidentiality, Integrity and Availability (CIA Triad), Real-world Applications of CIA. Threat and Risk Management: Threats, Vulnerabilities, Risks, Attack Surface, Data Breaches, Risk Assessment Basics. Human-Centric Security: Social Engineering, Phishing, Identity Theft, Insider Threats.	15	25

	Identity Protection: Authentication, Authorization, Password Security, Multi-Factor Authentication (MFA). Cryptography Fundamentals: Need for Encryption, Symmetric Encryption, Asymmetric Encryption, Digital Signatures.		
2	Data Privacy and Protection Data Privacy Fundamentals: Personal Data, Sensitive Personal Data, Personally Identifiable Information (PII), Data Privacy Principles. Data Lifecycle Security: Data Collection, Data Storage, Data Processing, Data Sharing, Data Disposal. Data Protection Techniques: Data Classification, Data Masking, Data Anonymization, Data Pseudonymization. Secure Data Handling: Secure File Sharing, Data Backup and Recovery, Secure Data Storage. Privacy Risks in Digital Ecosystems: Digital Footprint, Social Media Privacy, Data Privacy Challenges in Analytics. Organizational Data Protection: Data Protection Policies, Data Handling Best Practices, Data Loss Prevention Awareness.	15	25
3	Security Technologies for Data-Driven Systems Network and Communication Security: Network Security Fundamentals, Firewalls, VPN Concepts, Secure Communication. Cloud Security: Cloud Computing Fundamentals, Shared Responsibility Model, Cloud Security Challenges. Big Data Security: Security Challenges in Big Data Environments, Data Access Control, Data Security in Distributed Systems. AI and Security: Artificial Intelligence in Cybersecurity, AI-Assisted Threat Detection, Security Automation. Security of AI Systems: Data Poisoning, Model Security Risks, Adversarial Attacks (Conceptual Overview). Emerging Security Challenges: Deepfakes, Generative AI Security Risks, Future Trends in Information Security.	15	25
4	Data Governance, Compliance and Ethics Data Governance Fundamentals: Data Ownership, Data Stewardship, Data Governance Frameworks. Cyber Laws and Regulations: Information Technology Act 2000, Digital Personal Data Protection (DPDP) Act 2023. Data Privacy Compliance: Privacy Regulations, Consent Management, Compliance Requirements. Ethics in Data Science: Responsible Data Usage, Ethical Data Collection, Ethical AI Practices, Bias and Fairness in AI. Security Governance: Security Policies, Risk Management Fundamentals, Security Standards Overview. Industry Standards and Frameworks: ISO 27001 Overview, NIST Cybersecurity Framework Overview.	15	25
Total		60 Hours	100%

Course Outcomes:

CO No.	CO Statement	Blooms Taxonomy Level	Unit
CO-1	Explain the fundamental concepts, principles and challenges of information and data security.	U	1
CO-2	Apply data privacy principles and data protection techniques for securing information assets.	A	2
CO-3	Describe security technologies and their role in protecting modern data-driven systems.	U	3
CO-4	Analyze governance, compliance, legal and ethical requirements associated with data security and privacy.	AN	4

Teaching & Learning Pedagogy approaches:

- Problem - based Learning
- Case Study-Based Learning
- Discussion-Based Learning
- Industry-Oriented Learning
- Collaborative Learning
- Research-Based Learning
- Experiential Learning

List of Tutorials: **NA**

[illegible]

List of Experiments: NA

[illegible]

Major Equipment and Simulation Tools:

1. Modern Desktop PC/Laptop with Internet Connectivity
2. Multimedia Projector and Smart Classroom Facilities
3. Web Browser
4. OpenSSL Toolkit
5. Wireshark Network Analyzer
6. Python Environment (Anaconda/Jupyter Notebook)
7. Cloud Computing Platforms (AWS/Azure/GCP Educational Access)
8. Online Learning and Security Awareness Platforms
9. Data Privacy and Security Assessment Tools

Books Recommended(APA Format):

1. Whitman, M. E., & Mattord, H. J. (2022). Principles of Information Security (7th ed.). Cengage Learning.
2. Stallings, W., & Brown, L. (2024). Computer Security: Principles and Practice (5th ed.). Pearson Education.
3. Andress, J. (2023). The Basics of Information Security: Understanding the Fundamentals of InfoSec in Theory and Practice (3rd ed.). Syngress.
4. Vacca, J. R. (2020). Computer and Information Security Handbook (3rd ed.). Elsevier.
5. Chander, H., & Jain, V. (2023). Cyber Security and Cyber Laws. PHI Learning Pvt. Ltd.
6. Solove, D. J., & Schwartz, P. M. (2021). Information Privacy Law (7th ed.). Aspen Publishing.
7. Kelleher, J. D., Tierney, B., & Grzymala-Busse, J. (2020). Data Science and Security: Protecting Data in a Data-Driven World. Springer.
8. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press. (Selected Topics on AI Security and Adversarial Learning)
9. National Institute of Standards and Technology (NIST). (Latest Edition). Cybersecurity Framework (CSF) 2.0. NIST Publications.
10. Digital Personal Data Protection Act, 2023 (Government of India Reference Document).

Term Work Module:

40 HOURS

Objective: To strengthen conceptual understanding of information security, data protection, privacy, governance, and emerging security technologies through case studies, research activities, presentations, and practical investigations.

Term Work	Statement	Hours	CO No.
Term Work 1	Case Study Analysis of Major Data Breaches (Equifax, Facebook-Cambridge Analytica, Aadhaar, Marriott, etc.) focusing on threats, vulnerabilities, impact, and mitigation	10 Hours	CO1

	measures.		
Term Work 2	Study and Presentation on Data Privacy and Protection Practices in Digital Platforms (Google, Meta, Amazon, Microsoft, etc.) including privacy policies and user consent mechanisms.	10 Hours	CO2
Term Work 3	Comparative Analysis of Security Technologies used in Data-Driven Systems such as Encryption, Cloud Security Controls, Access Control Mechanisms, and AI-Assisted Security Solutions.	10 Hours	CO3
Term Work 4	Report and Presentation on Data Governance, DPDP Act 2023, Ethical Data Usage, and Responsible AI Practices using real-world scenarios.	10 Hours	CO4
Total Hours		40	

Self Learning Module:

20 HOURS

Objective: To encourage independent exploration of advanced concepts, industry practices, and emerging developments in information and data security.

Recommended Online Resources:

Sr. No.	Unit Name/Topic Name	Resource Link
1	Foundations of Information and Data Security Study of information security principles, threats, vulnerabilities, risk assessment, authentication, and cryptography fundamentals.	1. College Recorded Videos 2. NPTEL Courses 3. Information and Cyber Security https://www.youtube.com/playlist?list=PLYwpaL_SFmcArHtWmbs_vXX6soTK3WEJw 4. Information Security https://www.youtube.com/playlist?list=PL8DVBg0tfxv2elC0-4Y6l5ZDy1qHlWpQd 5. Information Security Foundations https://www.coursera.org
2	Data Privacy and Protection Study of PII, data privacy principles, data protection techniques, digital footprint, and privacy challenges in digital platforms.	1. College Recorded Videos 2. NPTEL Courses 3. Data Protection And Data Privacy https://cag.gov.in/uploads/media/Pursuit-9th-edition-0676a48f6b8c7a4-87699484-1-069098113f08f47-45386346.pdf 4. What Is Data Privacy in Cybersecurity? https://www.youtube.com/watch?v=bet

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3	Security Technologies for Data-Driven Systems Study of cloud security, big data security challenges, AI security risks, and emerging security technologies.	1. College Recorded Videos 2. NPTEL Courses 3. Google Cloud Cybersecurity Professional Certificate http://coursera.org/professional-certificates/google-cloud-cybersecurity-certificate 4. Data Security in the Age of AI https://www.youtube.com/watch?v=V5UXw5EwzZQ
4	Data Governance, Compliance and Ethics Study of DPDP Act 2023, IT Act 2000, data governance frameworks, compliance requirements, and ethical AI practices.	1. College Recorded Videos 2. NPTEL Courses 3. Privacy Law and Data Protection https://www.coursera.org 4. Certificate Course on DPDPA https://www.dpdpa.com/dpdpa-certificate-course.html

CO-PO-PSO Matrix:

Co. No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2
CO-1	3	2	-	-	-	-	2	2	-
CO-2	3	3	2	-	-	-	2	2	3
CO-3	2	3	2	2	-	-	2	2	3
CO-4	2	3	3	2	2	2	2	-	2

CO-PO-PSO MAPPING JUSTIFICATION FOR INFORMATION AND DATA SECURITY

CO No.	CO Statement	PO No.	Justification
CO1	Explain the fundamental concepts, principles and challenges of information and data security.	PO1	Understanding information security principles requires application of fundamental computing knowledge.
		PO2	Enables identification and analysis of security challenges, threats, and vulnerabilities.
		PO7	Encourages continuous learning of evolving security concepts and practices.

		PSO1	Supports understanding of security requirements while designing computing systems.
CO2	Apply data privacy principles and data protection techniques for securing information assets.	PO1	Requires understanding of data management and information security concepts.
		PO2	Involves identifying privacy risks and applying suitable protection mechanisms.
		PO3	Promotes analytical thinking for solving privacy and data protection challenges.
		PO7	Encourages independent learning of privacy regulations and protection techniques.
		PSO1	Supports secure handling and management of data-centric systems.
		PSO2	Involves the use of tools and techniques to solve real-world security problems.
CO3	Describe security technologies and their role in protecting modern data-driven systems.	PO1	Requires understanding of security technologies and their applications.
		PO2	Enables evaluation of appropriate security technologies for different environments.
		PO3	Promotes awareness of emerging technologies and innovative security solutions.
		PO4	Security technologies are applied across multidisciplinary domains.
		PO7	Encourages learning of evolving technologies and industry trends.
		PSO1	Supports analysis and design of secure computing systems.
		PSO2	Facilitates use of software tools and technologies for organizational security.

CO4	Analyze governance, compliance, legal and ethical requirements associated with data security and privacy.	PO1	Requires understanding of governance frameworks, standards, and regulations.
		PO2	Involves analysis of legal, ethical, and compliance requirements.
		PO3	Encourages ethical decision-making and responsible innovation.
		PO4	Governance activities require collaboration among technical, legal, and business stakeholders.
		PO5	Supports management and implementation of compliance initiatives.
		PO6	Requires effective communication through reports, documentation, and presentations.
		PO7	Encourages continuous learning of evolving laws and regulations.
		PSO2	Supports strategic implementation of governance and compliance practices in organizations.

MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS

SDG Mapping for Information and Data Security (DSM5061C)		
SDG No. & Title	Relevance to Course	Justification
SDG 4: Quality Education	High	Promotes digital literacy, cybersecurity awareness, privacy awareness, and responsible use of information systems.
SDG 9: Industry, Innovation and Infrastructure	High	Supports secure digital infrastructure, data protection, innovation, and adoption of trustworthy technologies.
SDG 16: Peace, Justice and Strong Institutions	High	Encourages ethical use of technology, data privacy, cybersecurity governance, and regulatory compliance.

SDG 17: Partnerships for the Goals	Moderate	Security governance and compliance require collaboration among organizations, regulators, and technology stakeholders.
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Explanation

The course promotes secure and responsible management of information assets, strengthens awareness of privacy and cybersecurity practices, and supports ethical and lawful use of data. It contributes to building secure digital ecosystems and prepares students to address security, governance, and compliance challenges in modern data-driven environments.